

BÀI TẬP TUẦN 6 - PHƯƠNG PHÁP NEWTON

5. Use Newton's method to find solutions accurate to within 10^{-4} for the following problems.

(a) $x^3 - 2x^2 - 5 = 0$, $[1, 4]$

(c) $x - \cos x = 0$, $[0, \pi/2]$

(b) $x^3 + 3x^2 - 1 = 0$, $[-3, -2]$

(d) $x - 0.8 - 0.2 \sin x = 0$, $[0, \pi/2]$

6. Use Newton's method to find solutions accurate to within 10^{-5} for the following problems.

(a) $e^x + 2^{-x} + 2 \cos x - 6 = 0$ for $1 \leq x \leq 2$

(b) $\ln(x - 1) + \cos(x - 1) = 0$ for $1.3 \leq x \leq 2$

(c) $2x \cos 2x - (x - 2)^2 = 0$ for $2 \leq x \leq 3$ and $3 \leq x \leq 4$

(d) $(x - 2)^2 - \ln x = 0$ for $1 \leq x \leq 2$ and $e \leq x \leq 4$

(e) $e^x - 3x^2 = 0$ for $0 \leq x \leq 1$ and $3 \leq x \leq 5$

(f) $\sin x - e^{-x} = 0$ for $0 \leq x \leq 1$, $3 \leq x \leq 4$ and $6 \leq x \leq 7$